

# From Raw Sensor Messages To Model-Ready Weather Data:

Lessons learned in a Real-Time Meteorological Pipeline



# What is the current scenario?

- **Rapid growth** of AI and NWP models
- Higher-resolution forecast **increases computational demand**
- Need to supply forecasting systems with **real-time data**
- **Climate change** is increasing the need to anticipate local and extreme events
- HPC enables scale, but **reliable data enables useful predictions**



# Why is meteorological data so difficult to work with?

- Reliance on meteorological **observations**
- Need to supply nowcasting and forecasting systems with **real-time** data
- Heterogeneous **sources** and changing data **formats**
- Limited **spatial** and **temporal** coverage
- Data **quality** and **missing data** due to sensors failures



# What are the main challenges?



## Data Acquisition

Each data source has a different format and reliability level.



## Data Processing

"One-size-fits-all" data treatments.



## Data Storage

Changes in weather stations are not tracked by storage tools.



## Forecast

Data Leakage and quality assurance from weather stations' data



## HPC / Operational Scalability

High-resolution models and real-time services require scalable compute, memory, and data movement.

# What challenges did we face at each stage?



## Data Acquisition

- Incomplete messages
- Missing expected variables
- Inconsistent values
- Missing data
- Need for temporal alignment



## Storage

- Transmission delays or gaps
- Changes in **message formats**
- Limited spatial coverage



## Data processing

- Robust imputation of missing data while preserving information
- Processing must adapt to data sources and forecast horizons

## What challenges did we face at each stage?



### Forecast

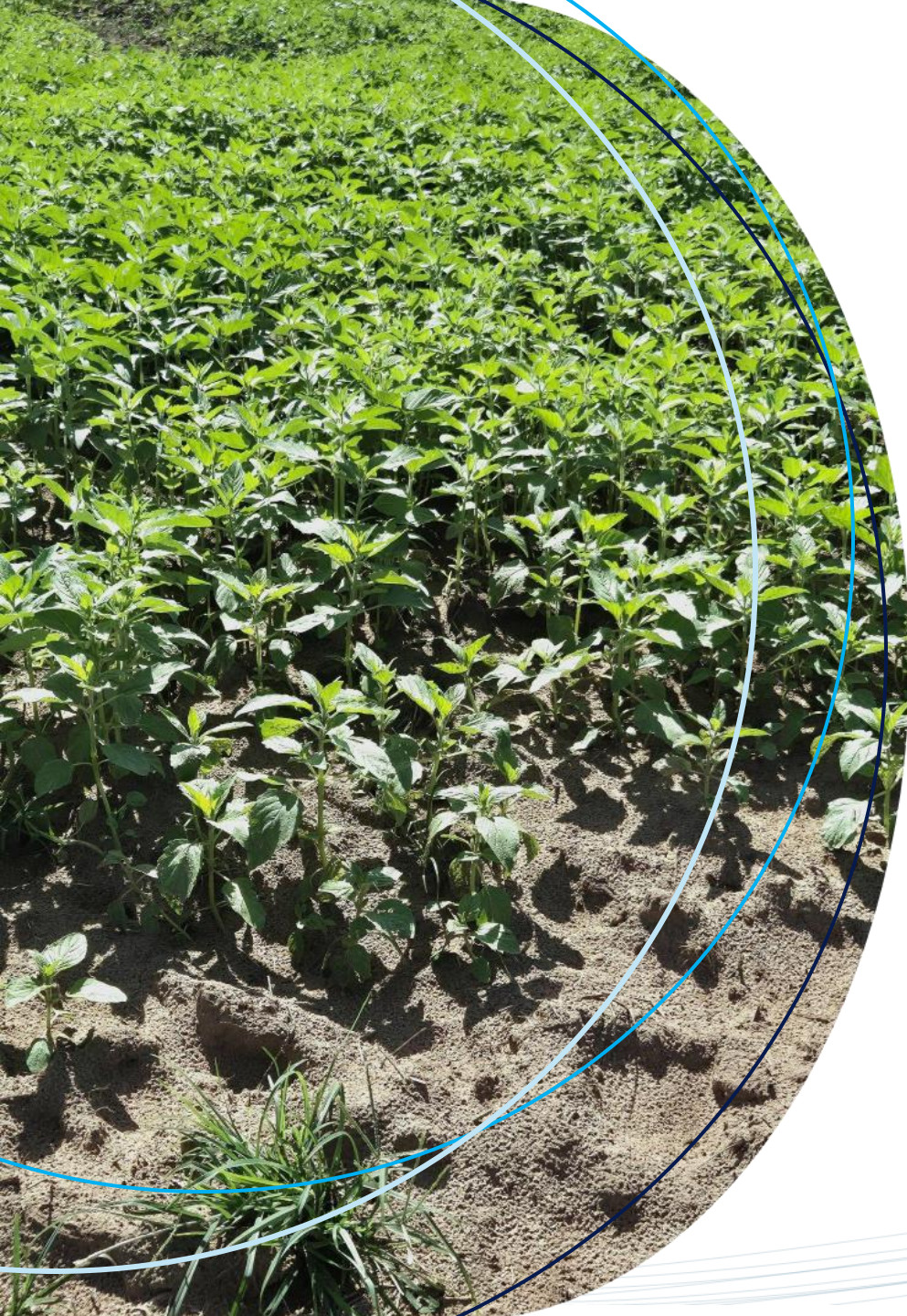
Some meteorological variables, such as precipitation, **remain difficult to predict even with robust models.**



### HPC and Operational scalability

**HPC must be integrated as a distributed, data-intensive system,** not as a conventional compute server.



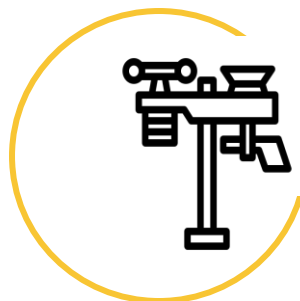


We are trying to build a data ecosystem...

**That provides trustworthy, reliable, interoperable and sustainable information.**



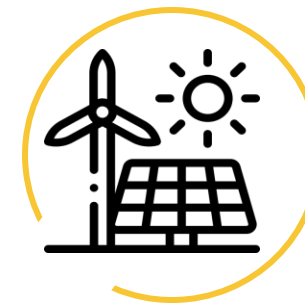
# Weather Stations



Initiatives based on **lower-cost and energy-efficient weather stations** can expand observational coverage



Low-complexity weather stations can improve maintainability and expand data coverage in remote areas.

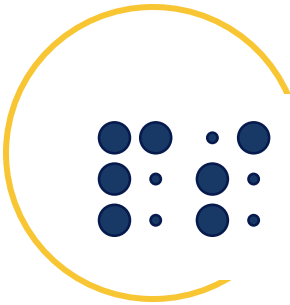


Complement existing high-precision stations by filling **spatial observation gaps.**



# Data Architecture

The solution uses a layered strategy inspired by the **Medallion Architecture**



## Bronze Layer

Raw data



## Silver Layer

Normalized and  
standardized data



## Gold Layer

Aggregated,  
domain-ready data

# Monitoring

## Managing history at scale

### Retention

What to preserve and for how long

### Partitioning

How to organize data

### Schema evolution

How to add or modify information without braking consumers

### Versioning

Preserve history and enable reproducibility

### Incremental processing

Process only what has changed

### Efficient querying

Keep queries fast as data volume grows

# What did we learn?



## Data Acquisition

- Weather station positioning
- land relief
- raw data storage



## Data Processing

Different variables require specific treatments.



## Data Storage

Storing large-scale historical data is challenging when future data requirements are uncertain.



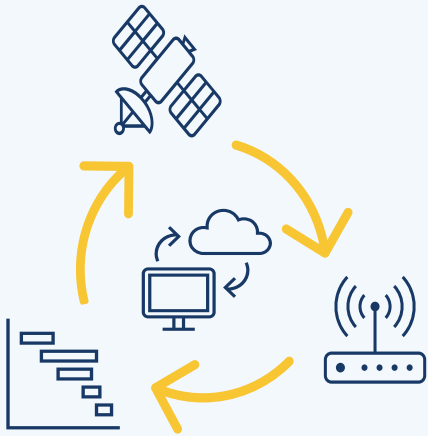
## Forecast

Different data pipelines for nowcasting and forecasting

*Land relief plays a big role in data precision, since the stations' positioning can impact the measurements.*



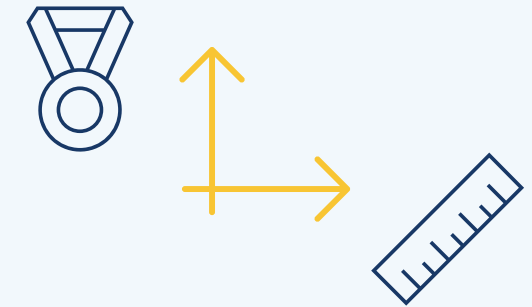
# How Do Lessons Learned Influence Weather Forecasting Challenges?



Interoperability is **essential**.



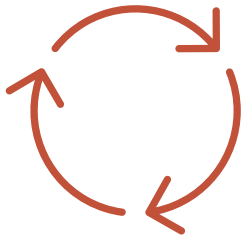
Observational networks have **different** manufacturers, protocols and heterogeneous data formats



For real-time forecasts, there is a permanent trade-off between **quality** and **latency**.

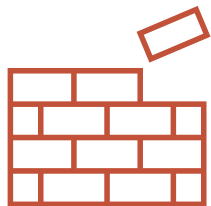
# HPC use for weather forecasting

## Next steps...



**Continuous retraining** is essential to ensure models remain accurate

*Weather patterns evolve and climate behavior changes over time*



Handling large-scale data processing and model training

*Consistent coverage is frequently prioritized because maintaining every observational instrument at optimal conditions is not always feasible.*

## Acknowledgments

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The Lenovo logo, consisting of the word "Lenovo" in white, lowercase letters on a red rectangular background.